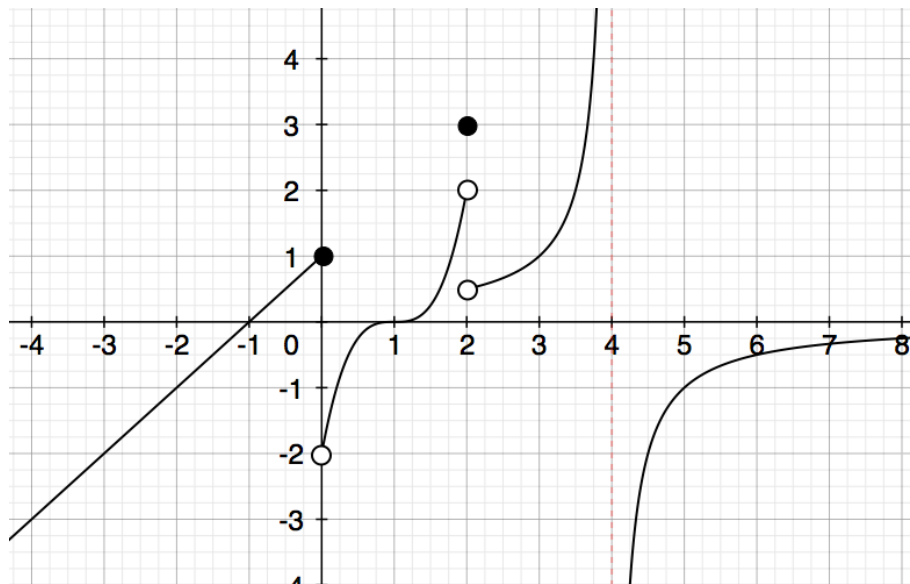


Module 2

Limits of functions and continuity

2.1 The concept of limit

(1) Consider the function $y = f(x)$ whose graph is depicted below.



Evaluate the following quantities, if they exist. If they do not exist, explain why.

- | | | |
|-------------------------------------|-----------------------------------|-------------------------------------|
| (a) $\lim_{x \rightarrow 0^-} f(x)$ | (b) $\lim_{x \rightarrow 0} f(x)$ | (c) $f(0)$ |
| (d) $\lim_{x \rightarrow 2^+} f(x)$ | (e) $f(2)$ | (f) $\lim_{x \rightarrow 2^-} f(x)$ |
| (g) $\lim_{x \rightarrow 4^+} f(x)$ | (h) $f(4)$ | (i) $\lim_{x \rightarrow 4} f(x)$ |

(2) Consider the following piecewise function

$$f(x) = \begin{cases} x^2, & x \geq 2 \\ 6 - x, & x < 2 \end{cases}$$

(a) Sketch the graph of $f(x)$.

(b) Evaluate the following quantities, if they exist. If they do not exist, explain why:

$$(a) \lim_{x \rightarrow 2^-} f(x) \quad (b) \lim_{x \rightarrow 2^+} f(x) \quad (c) \lim_{x \rightarrow 2} f(x) \quad (d) f(2).$$

(3) Consider the following piecewise function

$$f(x) = \begin{cases} 2x - 2, & x \leq 1 \\ x^3 - 1, & x > 1 \end{cases}$$

(a) Sketch the graph of $f(x)$.

(b) Evaluate the following quantities, if they exist. If they do not exist, explain why:

$$(a) \lim_{x \rightarrow 1} f(x) \quad (b) f(1)$$

2.2 Limit laws

(1) Use limit laws to evaluate the following limits. In your workings, clearly state which law you are using. Be very careful to conform to the notation discussed in lectures.

Important Note: You are expected to use the Study Guide for this question (Section 2.2). It is not required to memorise the number associated with each limit law.

$$\begin{array}{ll} (a) \lim_{x \rightarrow -1} (2x^4 - 3x^3 + 6x + 1) & (b) \lim_{x \rightarrow 1} \frac{2x^2 + 3}{x - 2} \\ (c) \lim_{x \rightarrow 2} 2(x^2 - 6)^2 & (d) \lim_{x \rightarrow 1} \left(\frac{1 - 2x}{2x^2 - x + 1} \right)^4 \end{array}$$

2.3 Limits at infinity and infinite limits

(1) Use limit laws to evaluate the following limits, if they exist. If they are not immediately computable, then indicate the indefinite form that arises. In your workings, clearly state which law you are using.

Important Note: Again, you are expected to use the Study Guide for this question (Section 2.3). It is not required to memorise the number associated with each limit law.

$$\begin{array}{ll} (a) \lim_{x \rightarrow -\infty} (2x^4 - x + 1) & (b) \lim_{x \rightarrow \infty} \frac{2x^2 + x - 3}{x + 7} \\ (c) \lim_{x \rightarrow 1^-} \frac{2}{(x - 1)^3} & (d) \lim_{x \rightarrow -3^+} \frac{x^2 - 1}{x + 3} \\ (e^*) \lim_{x \rightarrow \infty} \frac{x^2 - 3x + 6}{x - 10} & (f^*) \lim_{x \rightarrow 2} \left(\frac{x^2 + 1}{x^3 - 8} \right)^2 \end{array}$$

2.4 Simple cases for indefinite forms $\left(\frac{0}{0}\right)$, $\left(\frac{\infty}{\infty}\right)$ and $(\infty - \infty)$

(1) Use limit laws and the techniques of simplification, factorisation and rationalisation, where appropriate, to evaluate the following limits:

- | | |
|--|--|
| (a) $\lim_{x \rightarrow -3} \frac{x^2 - x - 12}{x + 3}$ | (b) $\lim_{x \rightarrow 2} \frac{x^2 + 3x - 10}{x^2 - x - 2}$ |
| (c) $\lim_{x \rightarrow 2} \frac{\sqrt{2x + 5} - 3}{x - 2}$ | (d) $\lim_{x \rightarrow 3} \frac{\sqrt{x + 1} - 2}{x^2 - x - 6}$ |
| (e) $\lim_{x \rightarrow \infty} \frac{3x^3 - 2x + 1}{x^3 - x^2 + 2x + 1}$ | (f) $\lim_{x \rightarrow -\infty} \frac{3x^2 + 6x - 1}{6x^3 - x + 10}$ |
| (g) $\lim_{x \rightarrow \infty} (\sqrt{x + 3} - \sqrt{x + 1})$ | (h*) $\lim_{x \rightarrow \infty} (\sqrt{x^2 - 3} - \sqrt{x^2 + x + 1})$ |
| (i) $\lim_{x \rightarrow -2} \frac{\frac{1}{2} + \frac{1}{x}}{2 + x}$ | (j*) $\lim_{x \rightarrow \infty} (\sqrt{9x^2 + 3x + 2} - 3x)$ |

Note: In questions (h) and (j), rationalisation alone is not sufficient to evaluate the limit correctly. Further factorisation of the numerator and denominator is also required after rationalisation.

2.5 The sandwich theorem and standard limits

(1) Using the sandwich theorem, evaluate the following limits

- | | |
|--|--|
| (a) $\lim_{x \rightarrow 0} x^3 \sin\left(\frac{7\pi}{x}\right)$ | (b) $\lim_{x \rightarrow \infty} e^{-x} \cos^2 x$ |
| (c) $\lim_{x \rightarrow \infty} \frac{\arctan(3x)}{\sqrt{x}}$ | (d) $\lim_{x \rightarrow 2^+} \sqrt{x - 2} \cdot \sin\left(\frac{x + 3}{x - 2}\right)$ |

(2) Use algebra, limit laws and standard limits SL. 1 - SL. 5 to evaluate the following expressions:

- | | |
|--|---|
| (a) $\lim_{x \rightarrow 0} \frac{\sin(2x)}{x}$ | (b) $\lim_{x \rightarrow 0} \frac{1 - \cos^2 x}{2x}$ |
| (c) $\lim_{x \rightarrow 0} \frac{\cos x - \sin^2 x - 1}{3x^2}$ | (d) $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{5x}$ |
| (e) $\lim_{x \rightarrow 0} \frac{e^{2x} - 2e^x + 1}{3x^2}$ | (f) $\lim_{x \rightarrow 0} \frac{\tan x}{7x}$ |
| (g) $\lim_{x \rightarrow 0} \frac{\ln(x^2 + 2x + 1) \ln(x + 1)}{4x^2}$ | (h) $\lim_{x \rightarrow 0} \frac{\ln(1 - x^2) - \ln(1 - x)}{2x}$ |
| (i*) $\lim_{x \rightarrow 0} \frac{\sqrt{1 + \sqrt{x + 1}} - \sqrt{2}}{x}$ | (j*) $\lim_{x \rightarrow 0} \frac{2 + x - 2\sqrt{1 + x}}{3x^2}$ |

2.6 Continuity: definition and examples

(1) Consider the following piecewise function:

$$f(x) = \begin{cases} 2x - 1, & x \neq 1 \\ 2, & x = 1. \end{cases}$$

(a) Sketch the graph of $f(x)$.

(b) Evaluate $\lim_{x \rightarrow 1^-} f(x)$.

(c) Evaluate $\lim_{x \rightarrow 1^+} f(x)$.

(d) Is $f(x)$ continuous in $x = 1$? If not, what type of discontinuity is present? Explain why.

(2) Consider the following piecewise function:

$$f(x) = \begin{cases} x + 2, & x < 0 \\ e^x, & 0 \leq x \leq 1 \\ \sqrt{x - 1} & x > 1 \end{cases}$$

(a) Sketch the graph of $f(x)$

(b) Evaluate $\lim_{x \rightarrow 1^-} f(x)$ and $\lim_{x \rightarrow 1^+} f(x)$.

(c) Evaluate $\lim_{x \rightarrow 0^-} f(x)$ and $\lim_{x \rightarrow 0^+} f(x)$.

(d) Is $f(x)$ continuous at $x = 0$ and $x = 1$? If not, what type(s) of discontinuities are present? Explain why.

(3) Consider the following piecewise function:

$$f(x) = \begin{cases} \frac{1}{(x-1)^2}, & x < 1 \\ \ln(x+1) & x > 1 \end{cases}$$

(a) Sketch the graph of $f(x)$

(b) What type of discontinuity is present at $x = 1$?

2.7 Theorems on continuity and applications to limits

(1) For the following functions, use theorems on continuity to find the largest possible sets on which they are continuous.

(a) $y = x^2 + \sqrt{3x - 2}$

(b) $y = \frac{2x + 7}{2x^2 - x + 1}$

$$\begin{array}{ll}
\text{(c)} & y = \sqrt{2 + |x|} - \sqrt{2 - |x|} \\
\text{(d)*} & y = \ln \left(\frac{x^2 - 2}{\sqrt{x - 1} - 1} \right) \\
\text{(e)} & y = \frac{\ln(1 - x^2)}{2^x - 3^x} \\
\text{(f)} & y = \exp \left(\frac{x^2 - 3x - 2}{x^2 - 3x + 2} \right) \\
\text{(g)*} & y = \frac{\arctan(x^2)}{\operatorname{arctanh}(x - 1)} \\
\text{(h)} & y = \frac{\ln(\arcsin x)}{\sinh(x - 2)}
\end{array}$$

(2) For all the examples (a)-(h) above, say whether the given functions are continuous on their natural domain.

(3) Find the value of $k \in \mathbb{R}$ such that

$$f(x) = \begin{cases} x^3 - kx + 8, & x \leq 1 \\ x^2 + 2kx + 2, & x > 1 \end{cases}$$

is continuous everywhere in $\mathbb{R}$.

(4*) Find the value of $k, p \in \mathbb{R}$ such that

$$f(x) = \begin{cases} x^2 + kx + k, & x < 0 \\ kx + 2p, & 0 \leq x \leq 1 \\ px + 3k + 1, & x > 1 \end{cases}$$

is continuous everywhere in $\mathbb{R}$.

(Hint: Enforce the continuity of $f(x)$ at $x = 0$ and $x = 1$, then solve the system of equations in p, k .)

(5) Using limit laws and the property of continuous functions with respect to limits, evaluate (where possible) the following expressions. In your working, clearly justify your answers.

$$\begin{array}{ll}
\text{(a)} & \lim_{x \rightarrow 0} \frac{\ln(x + \sqrt{x + 4})}{\cos(2\pi x^4)} \\
\text{(b)} & \lim_{x \rightarrow 0} \sin(\sin x + \pi) \\
\text{(c)} & \lim_{x \rightarrow 1} \exp \left(\frac{2x - 1}{x^2 + 4x - 6} \right) \\
\text{(d)} & \lim_{x \rightarrow 1} \frac{\sqrt[3]{\ln(2 - x^2)} + 2}{\arcsin(x^2 + x - 2)} \\
\text{(e)*} & \lim_{x \rightarrow -1} |x|^{2|x| - x^2} \\
\text{(f)} & \lim_{x \rightarrow 4} \sqrt[3]{\sqrt{x - 2} - \sqrt{4 + 3x}} \\
\text{(g)} & \lim_{x \rightarrow 2} \arctan(\sinh(x^2 - 4x + 4) - \cosh(1 + \sqrt{x - 2}))
\end{array}$$

2.8 The “race to infinity”: comparing “speeds” for the $\left(\frac{\infty}{\infty}\right)$ case

(1) Using algebraic manipulations and considerations about “speeds at infinity” of different functions, evaluate the following limits (if they exist). In your workings, clearly motivate your answers.

$$(a) \quad \lim_{x \rightarrow \infty} \frac{x^{13} - 2x^7 + 10x^2 + 1}{6x^{11} + 3x^8 + x^4 - 3}$$

$$(c) \quad \lim_{x \rightarrow \infty} \frac{\sin x - \cos x}{x^2 + 3 \ln(1 + x)}$$

$$(e) \quad \lim_{x \rightarrow \infty} \frac{x^{2x}}{(2x)^x}$$

$$(b) \quad \lim_{x \rightarrow \infty} \frac{x^{20} - 3x^5 + 3x - 2 + \sqrt{x^3 + 6} + e^{-3x}}{x + e^x}$$

$$(d) \quad \lim_{x \rightarrow \infty} \frac{\tan x}{\tanh x}$$

$$(f) \quad \lim_{x \rightarrow \infty} \frac{\ln(1 + e^x)}{x^2 + 3}$$

2.9 The substitution technique and further standard limits

(1) Use algebra, continuity, limit laws and standard limits SL. 6 - SL. 8 to evaluate the following expressions:

$$(a) \quad \lim_{x \rightarrow 0^+} (x^2 \ln x - x \ln(x^2))$$

$$(b) \quad \lim_{x \rightarrow 0^+} (x^x + (2x)^x - x^{-x})$$

$$(c) \quad \lim_{x \rightarrow 0^+} \frac{e^{1-x \ln x} + 1}{x^2 \ln^2 x + 2x \ln x + 1}$$

$$(d) \quad \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^{3x}$$

$$(e) \quad \lim_{x \rightarrow \infty} \left(\frac{1 + x^2 + 2x}{x^2}\right)^{2x}$$

$$(f^*) \quad \lim_{x \rightarrow 0^+} (2^{(x^x)} - (3^x)^x)$$

(2) Using the given suggestions, apply the substitution technique to evaluate the following limits, using standard limits SL. 1 - SL. 8. Note that, in some cases, some manipulation is also required to recast the expression into a known SL.

$$(a) \quad \lim_{x \rightarrow 0^+} (1 + 2x)^{1/x}, \quad t = 1/x$$

$$(b) \quad \lim_{x \rightarrow 0} \frac{\cos(\sin x) - 1}{\sin^2 x}, \quad t = \sin x$$

$$(c^*) \quad \lim_{x \rightarrow 0} \frac{\sqrt{\tan x + 1} - 1}{\sin x}, \quad t = \tan x$$

$$(d) \quad \lim_{x \rightarrow 0} \frac{\arcsin x}{x}, \quad t = \arcsin x$$

$$(e^*) \quad \lim_{x \rightarrow \infty} \left(1 + \frac{1}{\sinh x}\right)^{\tanh x}, \quad t = \sinh x$$

$$(f^*) \quad \lim_{x \rightarrow 0} \frac{(\exp(\sin x) - 1)^2}{\tan^2 x}, \quad t = \sin x$$

$$(g) \quad \lim_{x \rightarrow 0} (\cosh^2 x - 1)^{\sinh x}, \quad t = \sinh x$$

$$(h^*) \quad \lim_{x \rightarrow 0} \frac{\ln(2 - \cos x)}{\sin^2 x}, \quad t = 1 - \cos x$$